Goal: Interpret the definite integral as the (signed) area underneath the graph of a function.
Recall Left/Right Riemann sums (L_N and R_N) are used to approximate the (signed) area underneath the graph of a function.

Question: What happens when we take $N \to \infty$? See what happens with a Desmos presentation (link here).

$$\lim_{N\to\infty} L_N = \lim_{N\to\infty} R_n =$$

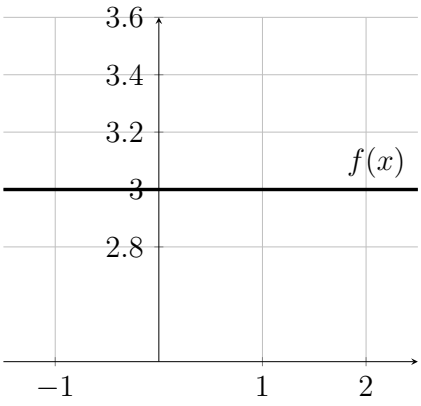
Definition. The **(definite) integral** of a function f from a to b is denoted as follows:

$$\int_a^b f(x)\,dx \overset{\text{def}}{=} \lim_{N\to\infty} L_N = \lim_{N\to\infty} R_n$$

$$=$$

a and b are called the upper and lower limits (or bounds) of integration, respectively. If the integral is finite, we say f is (Riemann) integrable on $[a, b]$.

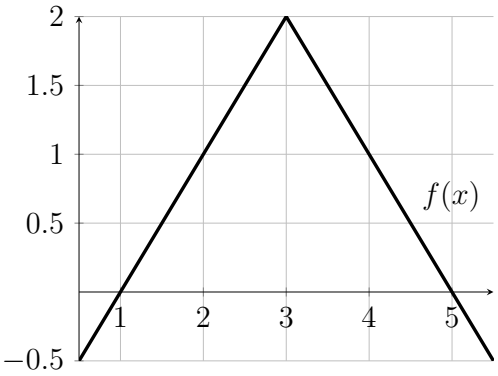
Example 1. Compute $\displaystyle\int_{-1}^{2} 3\,dx$



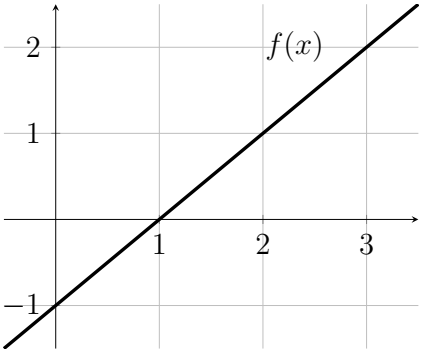
NOTE: In general, if k is a constant:

$$\int_a^b k \, dx =$$

Example 2. Compute $\int_1^5 (2 - |x - 3|) \, dx$

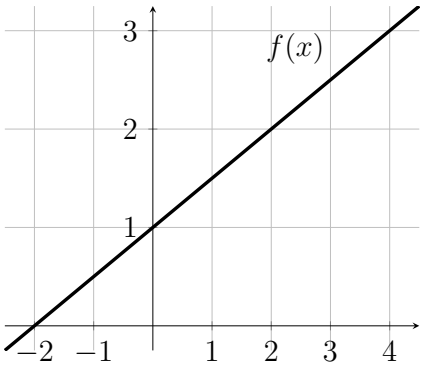


Example 3. Compute $\int_0^3 (x - 1) \, dx$

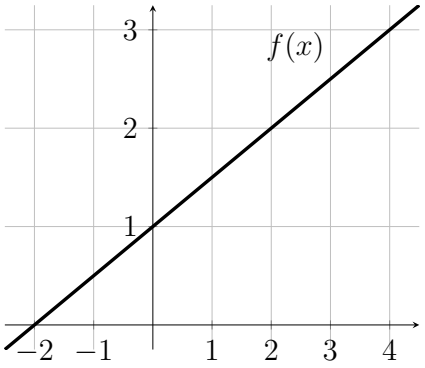


Example 4. Compute $\int_0^4 \left(\frac{1}{2}x + 1\right) \, dx$

(a) METHOD 1:



(b) METHOD 2:



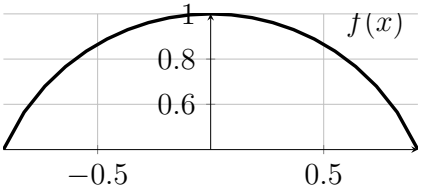
NOTE (Additivity of Integrals Example):
Rewriting METHOD 2 using integrals, we get:

$$\int_0^4 \left(\frac{1}{2}x + 1\right)\, dx =$$

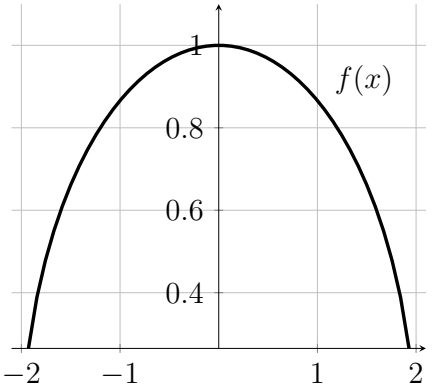
I.e.,

$$\int_{-2}^4 \left(\frac{1}{2}x + 1\right)\, dx =$$

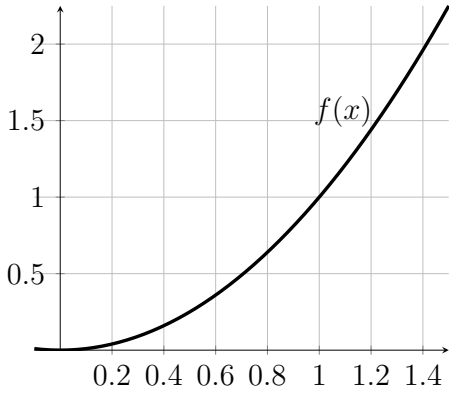
Example 5. Compute $\displaystyle\int_0^1 \sqrt{1 - x^2}\, dx$



Example 6. Compute $\displaystyle\int_{-2}^2 \sqrt{1 - \frac{x^2}{2^2}}\, dx$



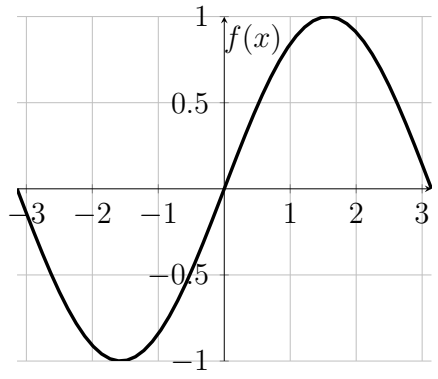
Example 7. Compute $\int_1^1 x^2 \, dx$



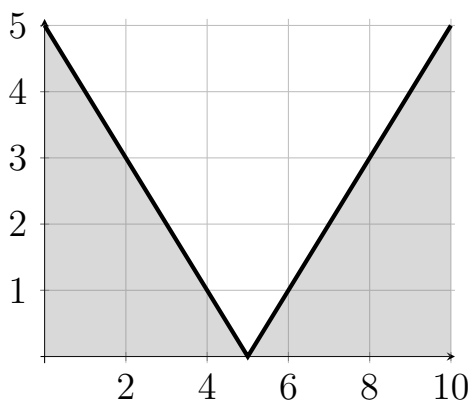
NOTE: In general,

$$\int_a^a f(x) =$$

Example 8. Compute $\int_{-\pi}^{\pi} \sin x \, dx$



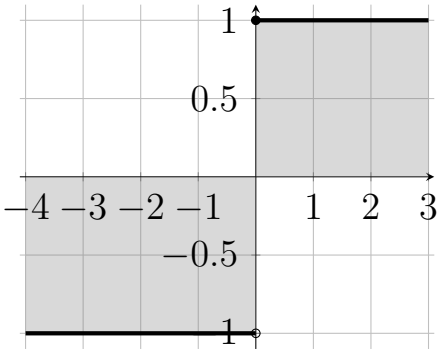
Example 9. Consider the shaded region below:



(a) Describe the area of the shaded region using an integral:

(b) Compute the area of the shaded region.

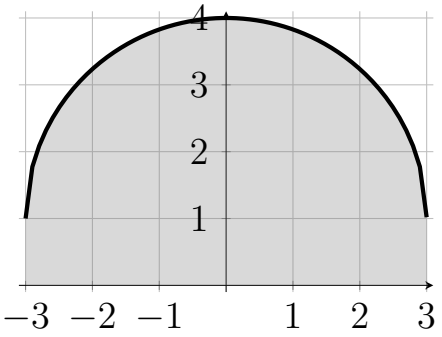
Example 10. Consider the shaded region below:



(a) Describe the area of the shaded region using an integral:

(b) Compute the area of the shaded region.

Example 11. Consider the shaded region below:



(a) Describe the area of the shaded region using an integral:

(b) Compute the area of the shaded region.

NOTE (Linearity of Integrals Example):
Rewriting our work using integrals, we see:

$$\int_{-3}^{3} (1 + \sqrt{9 - x^2})\, dx =$$